SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I YOGESH (192426035 9) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	1	1	1	1
Application	20%	1	1	2	1
Organization	15%	1	1	1	2
Accuracy	10%	1	1	1	1
Clarity	5%	1	1	1	1

Total Marks 30 **/40**

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of

3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organizatio n	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professiona l presentatio n	Understanda ble with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Computer Networks CO1 AT-1 (40) YOGIESH V 1924260359

30/40

1. A company needs to transfer a 150 mb design file from a client computer to a remote server over network of 50 mbps. Layer divided into

Given.

original file = 150mb
Network bandwith = 50 mbps
transport segment size = 1500 bytes
datalink layer overhead = 40 bytes
Assume 1MB = 1,000,000 = 150,000,000 bytes.

Step by step Calculation:

*, Convert the file size,

File Size = 150 × 1,000,000 = 150,000,000bytes

*, Calculate Segments:
The number of full segments is found by dividing the total file size.

Number of Segments = 150,000,000 ÷ 1500 = 100,000

* Calculate the no. of frames.

Assuming one transport layers Segment is the no. of frames

Number of frames = 100,000 frames

* Calculate Data link layer overhead

Every frame add 40 bytes of header and trailer information.

Total overhead = 100,000 x 40 = 4,000,000 bytes
= 4 MB

* Calculate total transmission data

Each frame contains 1500 bytes of segment data And 40 bytes of Data link information.

1 frame = 1540 frames

Total transmitted data = 100,000 x 1540 = 154,000,000 bytes
= 154 MB

* Estimated transmission time:

Bandwith is measured in bits per second, so total no of bytes must converted to bits.

Total bits = 154,000,000 x 8 = 1,232,000,000 bits

Transmission time = 1232,000,000 ÷ 50,000,000 = 24.64 Sec

Detailed Explianation:-

The transport layer provides communication between running on different computers, TCP, for example, assigns Sequence number to data, acknowledge recived bytes dectes missing information, and retransmit data when nessary.

These layers perform different complementary task. The transport layer fouces on end-end communications, whereas the data link is concerned mainly with delivery across a singe network layer.

2 transport layer protocol uses a sliding window Size of frames. Each frame carries 1024 bytes of user data. If the Sender trasmit 48 frames. Calculate the number of transmitted, determine the total number of retransmission.

Given,

Silding window frames: 6 frames
user data in each frame = 1024 bytes
total frames to be sent = 48 frames

Step 1:

Determine how many frames in one window

frames per window = 6

Step 2:

Calculate the no of transmission windows

$$\Rightarrow \frac{48}{6}$$

$\Rightarrow$ 8 windows

Step 3:

Calculate the number of retransmission

$\Rightarrow 8 \times 1$

$\Rightarrow$ 8 frames

Step 4:

Calculate the amount of data

$= 48 \times 1024$

$= 49,152$ bytes

The calculation gives eight transmission window the original user data is 49152 bytes. Sliding window improves efficiency by allowing multiple frames to be in transit, while flow control perfects the recives error on the protocol window rules.

3. A 12,000-byte IP packet is transmitted through a router where maximum Transmission unit (MTU) is 1500 bytes. Assume each fragment created requires 20-byte IP header. Calculate total number of fragments created and total header overhead introduced. Explain how the Network layer ensure delivery of packets

Given,

Original IP packet data to be fragment:

=> 12000 bytes

HTU = 1500 bytes

IPv_4 = header per fragments = 20 bytes

Step 1:

Calculate the max payload per fragment

=> 1500 - 20

=> 1480 bytes

Step 2:

Calculate the no of fragments

$$\Rightarrow \frac{12000}{1480}$$

$$\Rightarrow 8.10$$

full fragments carry,

=> 8 × 1480

=> 11 840

emaing Payload = 12000 - 11840

=> 168 bytes

Total fragments = 8 + 1

= 9 fragments required

Step 3:

Calculate the total IP header overload:

=> 9 x 10

=> 180 bytes

Step 4:

Calculate the total transmitted IP layer.

=> 12000 + 180

=> 12,180 bytes

5. A multimedia application transfers video file. The Session layer insert Synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs 390 MB has been transfered. Determined the number of checkpoints.

Given,

Total video file = 600 MB
Checkpoint interval = 60 MB
Failure at => 390 MB

Step 1:

Checkpoint position

=> *60 MB
*,120 MB
*,180 MB
*,240 MB
*,300 MB
*,360 MB
*,420 MB
*,480 MB
*,540 MB
*,600 MB

Step 2:

Calculate the no of checkpoints before failure

$$\Rightarrow \frac{390}{60}$$

$$\Rightarrow 6.5$$

$$\Rightarrow 6 \text{ checkpoints}$$

Step 3:

Calculate data that must be resumbitted

$$\Rightarrow 390 - 360$$

$$\Rightarrow 30 \text{ MB should be retransmitted}$$

Step 4:

Remaining data

$$\Rightarrow 600 - 390$$

$$\Rightarrow 210 \text{ MB}$$

Synchronization points preserve a record of progress in a session. When a failure occurs the applications can resume from a known state rather than repeating a whole operation.